

Computing the SVD by Hand

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Linear transformations and matrices ([Note 500](#)); Eigenvectors and eigenvalues ([Note 530](#)).
- **Leads to:** Low-rank approximation (truncated SVD) ([Note 612](#)); Latent semantic analysis ([Note 613](#)); Moore-Penrose pseudo-inverse ([Note 613](#)).
- **Compare with:** Eigenvectors and eigenvalues ([Note 530](#)).

1. Overview

Key point: Multiplying A by its own transpose makes one of the two orthogonal matrices disappear. The eigenvectors of $A^T A$ are the right singular vectors, its eigenvalues are the squared singular values, and $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ gives the left singular vectors.

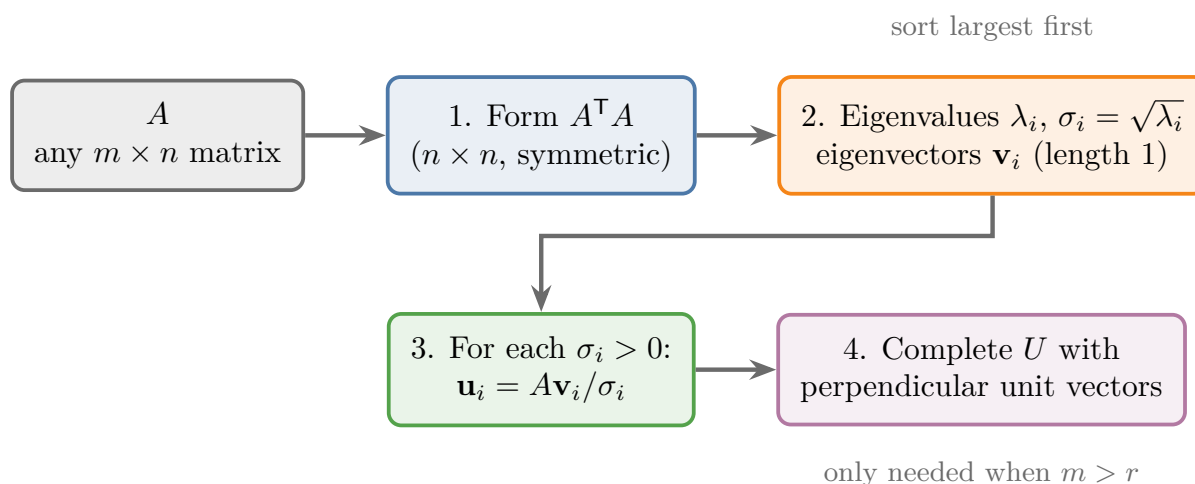


Figure 1: The four steps of computing an SVD by hand

The first SVD Note, the [SVD geometry Note](#) showed what $A = U\Sigma V^T$ means: rotate, stretch, rotate. It gave the factors of

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$

without saying where they came from. This Note finds them:

- why $A^T A$ and AA^T hold the answer (Section 2);
- the four-step recipe of Figure 1, worked on A (Section 3);

- a sign trap, and how $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ avoids it (Section 4);
- a matrix of rank 1, where a singular value is 0 (Section 5);
- the four subspaces that the SVD gives bases for (Section 6);
- a matrix that is not square (Section 7);
- why computers do not use this recipe (Section 8).

Finding eigenvalues and eigenvectors by hand, with $\det(A - \lambda I) = 0$, is taught in the [eigenvectors and eigenvalues Note](#); we use it here without repeating it.

2. Making one orthogonal matrix disappear

Key point: In $A^T A$ the two copies of U meet as $U^T U = I$ and vanish, leaving an eigen-decomposition of a symmetric matrix.

2.1 $A^T A$ gives V and the singular values

Key point: $A^T A = V \Sigma^T \Sigma V^T$: its eigenvectors are the $\mathbf{v}_i$, its eigenvalues are the σ_i^2 .

$A = U \Sigma V^T$ has two unknown orthogonal matrices. Finding both at once is hard, so we look for an expression in which one of them disappears.

1. **In words:** transpose A (the transpose of a product is the product of the transposes in reverse order), multiply by A , and use $U^T U = I$ (see the [SVD geometry Note](#), section 2.5).
2. **Formula:**

$$A^T A = (V \Sigma^T U^T)(U \Sigma V^T) = V \Sigma^T \Sigma V^T = V \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{bmatrix} V^T$$

3. **Example:** for A with rows $[3, 0]$ and $[4, 5]$,

$$A^T A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$

The right side of the formula is exactly an eigen-decomposition PDP^{-1} (see the [eigenvectors and eigenvalues Note](#), section 6.2), with $P = V$ and $P^{-1} = V^T$. So:

- the right singular vectors $\mathbf{v}_i$ are the eigenvectors of $A^T A$;
- the squared singular values σ_i^2 are its eigenvalues, so $\sigma_i = \sqrt{\lambda_i}$.

2.2 Why this always works

Key point: $A^T A$ is symmetric with eigenvalues ≥ 0 , so it always has orthonormal eigenvectors and real square roots.

A itself may have no useful eigenvectors, but $A^T A$ always does, for two reasons.

- $A^T A$ is **symmetric**. Its transpose is $A^T(A^T)^T = A^T A$, itself. A symmetric matrix always has a full set of perpendicular eigenvectors (the same fact that makes PCA work on a covariance matrix, see the [PCA step by step Note](#), section 4.3). So V can always be built as an orthogonal matrix.
- **Its eigenvalues are never negative**. For a unit eigenvector $\mathbf{v}$ with eigenvalue λ , $\lambda = \mathbf{v}^T A^T A \mathbf{v} = \|A\mathbf{v}\|^2 \geq 0$. So the square root $\sigma = \sqrt{\lambda}$ is a real, non-negative number. The formula also says σ is the length of $A\mathbf{v}$, as in the geometry Note.

Figure 2 shows where these numbers live for A with rows $[3, 0]$ and $[4, 5]$. A unit vector $\mathbf{v}$ goes round the circle, $A\mathbf{v}$ goes round an ellipse, and the graph tracks the length $\|A\mathbf{v}\|$. Watch the peak and the trough: the longest stretch, $\sqrt{45} = 6.71$, comes at $\mathbf{v}_1 = [1, 1]/\sqrt{2}$, and the shortest, $\sqrt{5} = 2.24$, at $\mathbf{v}_2 = [-1, 1]/\sqrt{2}$, exactly the eigenvectors and the square roots of the eigenvalues that Section 3 computes.

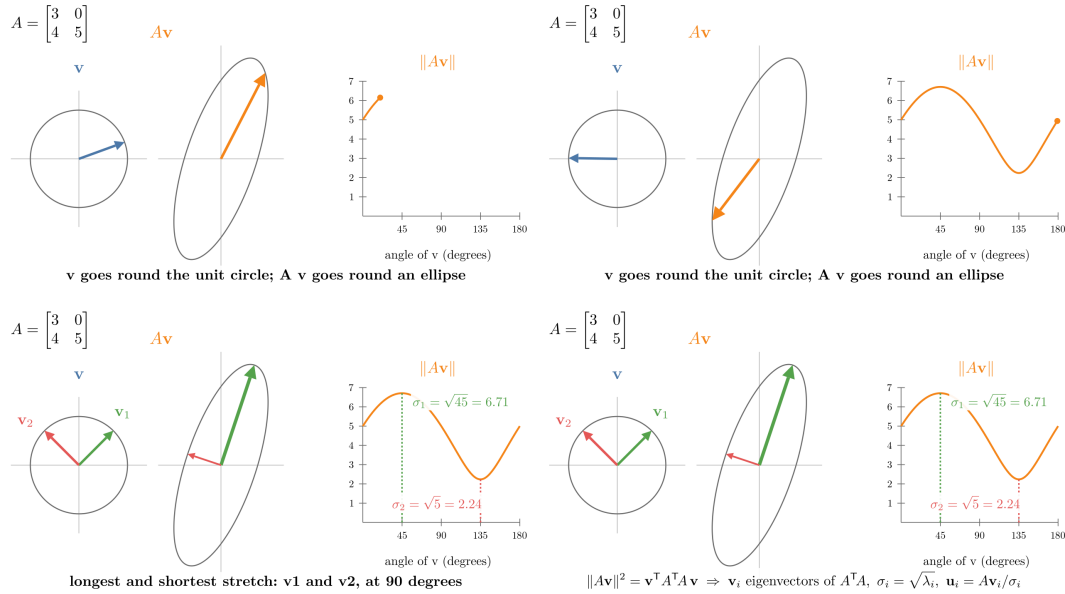


Figure 2: A unit vector $\mathbf{v}$ sweeps the circle and $A\mathbf{v}$ sweeps an ellipse; the length of $A\mathbf{v}$ peaks at $\sigma_1 = \sqrt{45}$ along $\mathbf{v}_1$ and bottoms out at $\sigma_2 = \sqrt{5}$ along $\mathbf{v}_2$, at 90° to $\mathbf{v}_1$

So $A^T A$ is a **symmetric matrix** (G-1932) and **positive semi-definite** (G-1532), for every matrix A of any shape.

2.3 AA^T gives U , with the same numbers

Key point: Multiplying in the other order removes V instead: $AA^T = U\Sigma\Sigma^T U^T$.

By the same steps, $AA^T = U\Sigma V^T V \Sigma^T U^T = U\Sigma\Sigma^T U^T$. So the left singular vectors $\mathbf{u}_i$ are the eigenvectors of AA^T , and its non-zero eigenvalues are the same σ_i^2 .

For A : AA^T has rows $[9, 12]$ and $[12, 41]$, a different matrix from $A^T A$. Yet both have the eigenvalues 45 and 5. The match is no accident: AB and BA always share their non-zero

eigenvalues.

3. The recipe, worked on a 2×2 matrix

Key point: Eigen-decompose $A^T A$, take square roots, then get each $\mathbf{u}_i$ by applying A to $\mathbf{v}_i$ and dividing by σ_i .

Figure 1 shows the four steps. We follow them for A with rows $[3, 0]$ and $[4, 5]$.

Step 1: form $A^T A$. As in Section 2.1, it has rows $[25, 20]$ and $[20, 25]$.

Step 2: eigenvalues and eigenvectors of $A^T A$.

1. **In words:** subtract λ from the diagonal and set the determinant to zero; then find the line of vectors each eigenvalue squishes to zero, and scale to length 1.
2. **Formula:**

$$\det(A^T A - \lambda I) = 0, \quad \sigma_i = \sqrt{\lambda_i}$$

3. **Example:**

$$\det \begin{bmatrix} 25 - \lambda & 20 \\ 20 & 25 - \lambda \end{bmatrix} = (25 - \lambda)^2 - 400 = 0 \implies 25 - \lambda = \pm 20$$

so $\lambda_1 = 45$ and $\lambda_2 = 5$ (largest first), and

$$\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.708, \quad \sigma_2 = \sqrt{5} \approx 2.236$$

For $\lambda_1 = 45$, $A^T A - 45I$ has rows $[-20, 20]$ and $[20, -20]$, which sends $[x, y]$ to zero when $x = y$. For $\lambda_2 = 5$, $A^T A - 5I$ has rows $[20, 20]$ and $[20, 20]$, which needs $y = -x$. Scaled to length 1:

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

The two vectors are perpendicular, as Section 2.2 promised.

Step 3: $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$. The step is the singular value equation $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$ solved for $\mathbf{u}_i$.

1. **In words:** apply A to each right singular vector and divide by its singular value. The result is automatically a unit vector.
2. **Formula:**

$$\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$$

3. **Example:**

$$\mathbf{u}_1 = \frac{1}{3\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

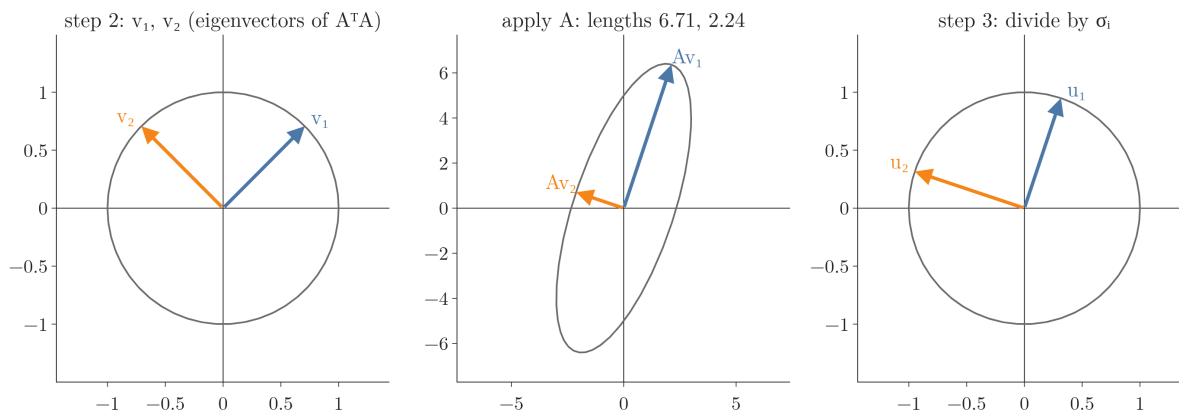


Figure 3: Steps 2 and 3 for A : the eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ of $A^T A$ (left), their images under A with lengths σ_1, σ_2 (middle), and the same images shrunk back to length 1, which are $\mathbf{u}_1, \mathbf{u}_2$ (right)

In Figure 3, watch the middle panel: $A\mathbf{v}_1$ and $A\mathbf{v}_2$ already point along the ellipse's axes, so dividing by their lengths only shrinks them back onto the circle. The $\mathbf{u}$'s come out perpendicular without any extra work. In symbols: $(A\mathbf{v}_1)^T(A\mathbf{v}_2) = \mathbf{v}_1^T(A^T A \mathbf{v}_2) = \lambda_2 \mathbf{v}_1^T \mathbf{v}_2 = 0$.

Step 4: complete U . Here there are already two $\mathbf{u}$'s for a 2×2 matrix, so nothing is missing. Sections 5 and 7 need this step.

Check. Multiplying the factors back together:

$$U\Sigma = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -3 \\ 9 & 1 \end{bmatrix}$$

$$U\Sigma V^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -3 \\ 9 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = A$$

Python: Checking each step with NumPy.

```
import numpy as np
```

```
A = np.array([[3, 0], [4, 5]])
lam, V = np.linalg.eigh(A.T @ A)    # eigh: for symmetric
lam, V = lam[::-1], V[:, ::-1]     # largest first
sigma = np.sqrt(lam)               # [6.708, 2.236]
U = A @ V / sigma                  # column i = A v_i / sigma_i
U @ np.diag(sigma) @ V.T           # gives back A
A @ V / sigma divides column i of AV by sigma_i: NumPy matches the 1D array sigma with
the columns.
```

4. The sign trap

Key point: Finding U and V separately as eigenvectors can pick mismatched signs and give a wrong product; computing $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ always matches them.

Section 2.3 suggests a shortcut: take the $\mathbf{u}$'s as the eigenvectors of AA^T , found on their own. For A , AA^T has rows $[9, 12]$ and $[12, 41]$, with eigenvalues 45 and 5. Its eigenvectors are the lines through $[1, 3]$ and through $[3, -1]$.

Every non-zero multiple of an eigenvector is an eigenvector too, so $\frac{1}{\sqrt{10}}[3, -1]$ is as correct an eigenvector as $\frac{1}{\sqrt{10}}[-3, 1]$. But it is the negative of the $\mathbf{u}_2$ that belongs with our $\mathbf{v}_2$. Using it gives

$$\frac{1}{\sqrt{10}} \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 5 & 4 \end{bmatrix} \neq A$$

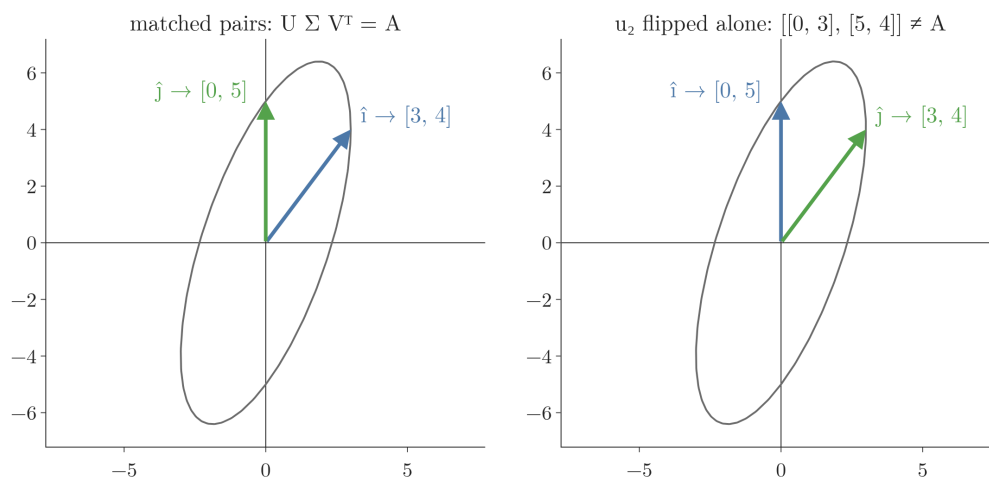


Figure 4: Left: the matched factors rebuild A . Right: the same factors with only $\mathbf{u}_2$ flipped. The ellipse is identical, but $\hat{i}$ and $\hat{j}$ land in swapped places

Each piece is a valid eigenvector, yet the product is a different matrix. Figure 4 shows why the mistake is easy to miss: both products have the singular values 6.71 and 2.24, so they draw the same ellipse; only where each input lands has changed. The pairs $(\mathbf{u}_i, \mathbf{v}_i)$ must match: flipping both signs is allowed, flipping one is not (see the [SVD geometry Note](#), section 5.1). The eigenvalue problem for AA^T knows nothing about V , so it cannot keep the pairs matched.

The fix is Step 3: compute $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ from the $\mathbf{v}_i$ already chosen. Then $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$ holds by construction, whatever sign each $\mathbf{v}_i$ was given.

5. A matrix of rank 1

Key point: A rank-1 matrix has one non-zero singular value; the zero one has no $\mathbf{u}$ from $A\mathbf{v}/\sigma$, so we complete U with a perpendicular unit vector.

Some matrices squash the plane flat. Take

$$C = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

Its columns $[2, 4]$ and $[1, 2]$ both lie on the line through $[1, 2]$. Every output $C\mathbf{x}$ is a mix of the two columns, so every output lands on that one line: C squashes the whole plane onto it.

- The set of all outputs is the **column space** (G-414) of C : here, the line through $[1, 2]$.
- The number of dimensions of the column space is the **rank** (G-1627) of the matrix: here 1. (The second row is twice the first, the algebraic sign of the same fact.)
- While the plane squashes onto a line, a whole other line of inputs, the one through $[-1, 2]$, lands on the origin. The set of inputs that land on $\mathbf{0}$ is the **null space** (G-1362).

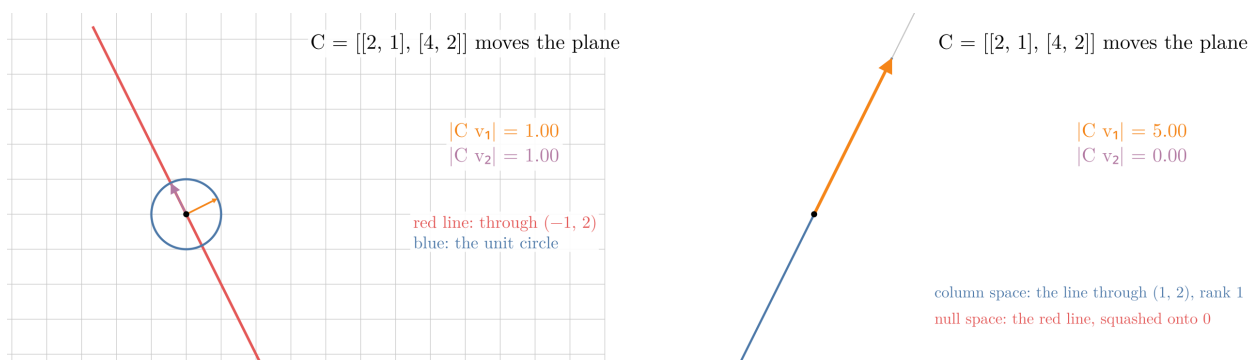


Figure 5: The rank-1 matrix C moves the grid: the whole plane, and the unit circle (blue) with it, squashes onto the line through $[1, 2]$, while every point of the red line through $[-1, 2]$ lands on the origin. $\mathbf{v}_1$ (orange) ends at length 5 and $\mathbf{v}_2$ (purple) at length 0. Idea after 3Blue1Brown, “Inverse matrices, column space and null space | Chapter 7, Essence of linear algebra”

In Figure 5, watch the red line while the grid moves: it shrinks towards the origin and vanishes, while the circle flattens into a segment. The two readouts end at 5 and 0, the two singular values the recipe now computes.

Step 1.

$$C^T C = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix}$$

Step 2. Its determinant is $20 \times 5 - 10 \times 10 = 0$, so one eigenvalue is 0. The eigenvalues of

a 2×2 matrix add up to its diagonal sum, $20 + 5 = 25$, so the other is 25. So $\sigma_1 = 5$ and $\sigma_2 = 0$, and the eigenvectors are

$$\mathbf{v}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$\mathbf{v}_2$ is the direction C squishes to nothing: $C[-1, 2] = [-2 + 2, -4 + 4] = [0, 0]$.

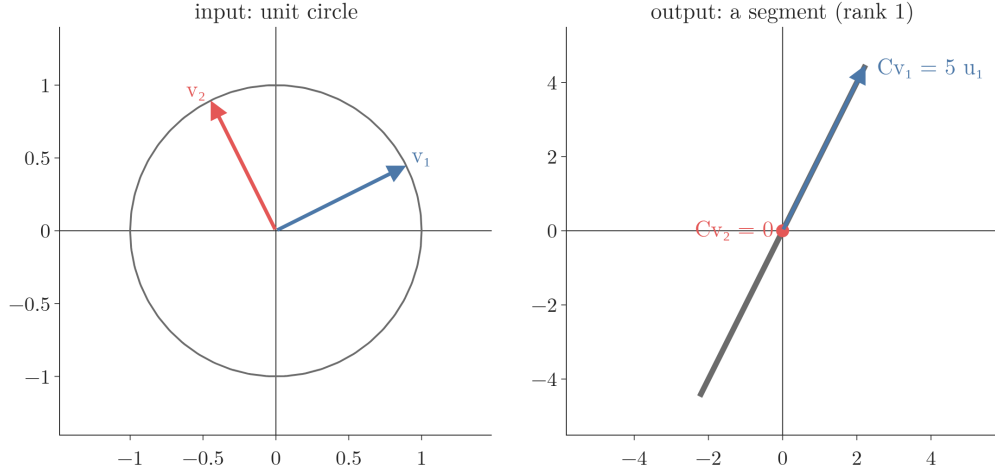


Figure 6: C flattens the unit circle onto a segment of the line through $[1, 2]$: $\mathbf{v}_1$ lands at length 5, $\mathbf{v}_2$ lands on the origin

In Figure 6, the ellipse of the 2×2 case has collapsed: its short axis σ_2 is 0, so the whole circle lands on one line, reaching 5 on each side of the origin.

Step 3. Only σ_1 is non-zero:

$$\mathbf{u}_1 = \frac{1}{5} C \mathbf{v}_1 = \frac{1}{5\sqrt{5}} \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Step 4. $\mathbf{u}_2 = C \mathbf{v}_2 / \sigma_2$ would divide zero by zero. Any unit vector perpendicular to $\mathbf{u}_1$ completes U to an orthogonal matrix; we take $\mathbf{u}_2 = \frac{1}{\sqrt{5}}[-2, 1]$. This $\mathbf{u}_2$ is multiplied by $\sigma_2 = 0$, so it never affects the product.

Result.

$$C = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

Only the first column of U and the first row of V^T meet a non-zero number, so the product shrinks to

$$C = 5 \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

A column times a row is a whole matrix of rank 1. The [low-rank approximation Note](#) builds every matrix out of such pieces.

Python: NumPy reports the zero singular value.

```
C = np.array([[2, 1], [4, 2]])
np.linalg.svd(C)[1]      # [5., 0.]
np.linalg.matrix_rank(C) # 1
```

`matrix_rank` from the [linear combinations, span and basis Note](#) works exactly this way: it counts the singular values that are not (almost) zero.

6. The four subspaces

Key point: The $\mathbf{v}$'s split the input space into the row space and the null space; the $\mathbf{u}$'s split the output space into the column space and the left null space. The SVD gives perpendicular bases for all four.

The rank-1 example shows a pattern that holds for every matrix. Figure 7 draws it for C .

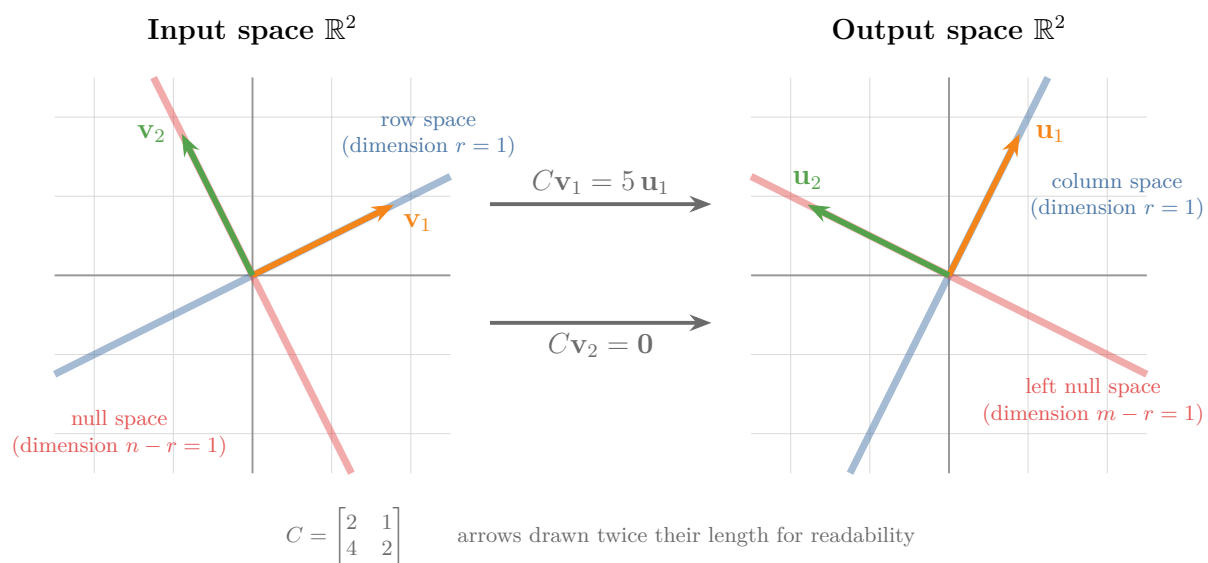


Figure 7: The SVD of C gives a basis for each of its four subspaces

Together they are the **four fundamental subspaces** (G-799). For an $m \times n$ matrix of rank r (there are r non-zero singular values):

Subspace	Where	Basis from the SVD	Dimension	For C
Column space (G-414) (span of the columns: every output)	output $\mathbb{R}^m$	$\mathbf{u}_1, \dots, \mathbf{u}_r$	r	line through $[1, 2]$
Null space (G-1362) (vectors sent to $\mathbf{0}$)	input $\mathbb{R}^n$	$\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$	$n - r$	line through $[-1, 2]$
Row space (G-1713) (span of the rows)	input $\mathbb{R}^n$	$\mathbf{v}_1, \dots, \mathbf{v}_r$	r	line through $[2, 1]$
Left null space (G-1078) (outputs never reached, perpendicular to them)	output $\mathbb{R}^m$	$\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$	$m - r$	line through $[-2, 1]$

The first two are the ones Figure 5 showed moving: the column space is where every output lands (the span of the columns from the [linear transformations and matrices Note](#), section 6), and the null space is everything the matrix squishes to the origin. The other two are their partners: the row space is the input directions perpendicular to the null space, and the left null space is the output directions perpendicular to the column space.

What makes the SVD's bases special is that they line up in pairs: A sends $\mathbf{v}_1$ to a multiple of $\mathbf{u}_1$, $\mathbf{v}_2$ to a multiple of $\mathbf{u}_2$, and so on, with no mixing. The null-space $\mathbf{v}$'s go to zero, which is where the zeros on the diagonal of Σ come from. Any method of making perpendicular bases could give bases of the four subspaces; only the SVD's bases also make the matrix diagonal.

7. A matrix that is not square

Key point: For a tall matrix, $A^\top A$ is small; the extra columns of U come from completing it with perpendicular unit vectors.

Take the 3×2 matrix B of the [SVD geometry Note](#) (section 6):

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Steps 1 and 2. $B^\top B$ is only 2×2 :

$$B^T B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad (2 - \lambda)^2 - 1 = 0 \implies \lambda = 3, 1$$

So $\sigma_1 = \sqrt{3} \approx 1.732$ and $\sigma_2 = 1$, with $\mathbf{v}_1 = \frac{1}{\sqrt{2}}[1, 1]$ and $\mathbf{v}_2 = \frac{1}{\sqrt{2}}[1, -1]$.

Step 3.

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}} B \mathbf{v}_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 0.816 \\ 0.408 \\ 0.408 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{1} B \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

Step 4. U is 3×3 and needs a third column, perpendicular to both. $\mathbf{u}_3 = \frac{1}{\sqrt{3}}[1, -1, -1]$ works: its dot products with $[2, 1, 1]$ and with $[0, -1, 1]$ are both 0. The vector $\mathbf{u}_3$ spans the left null space, the one direction in 3D that B never reaches.

The other route, eigenvectors of BB^T , would mean a 3×3 eigenvalue problem (eigenvalues 3, 1 and 0), and would still risk the sign trap of Section 4. For a tall data matrix with thousands of rows, $A^T A$ is the small one, so we always start from it.

8. How computers compute the SVD

Key point: Forming $A^T A$ squares the singular values, and small ones get lost in rounding. Libraries compute the SVD directly from A .

The recipe is for understanding and for small examples. On a computer it has a flaw. Numbers are stored with about 16 significant digits. Squaring a singular value of 10^{-9} gives 10^{-18} ; next to an eigenvalue of 4, that is far below the 16th digit and disappears in rounding.

Python: A nearly rank-1 matrix: its second singular value is tiny but not zero.

```
M = np.array([[1, 1], [1, 1 + 1e-9]])
np.linalg.svd(M)[1] # [2.0, 5.0e-10]: correct
np.sqrt(np.linalg.eigvalsh(M.T @ M))
# [0.0, 2.0]: the small one is lost
```

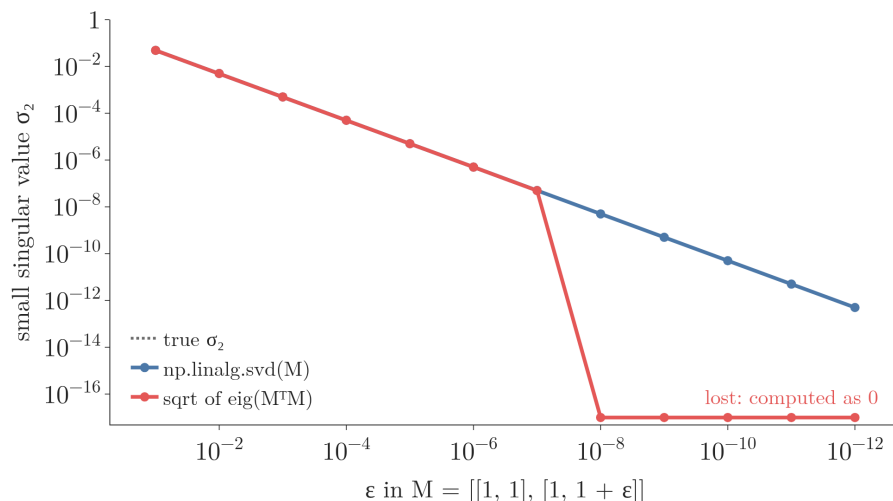


Figure 8: The small singular value of M as ε shrinks: `np.linalg.svd` (blue) follows the true value (dotted, under the blue line); the square root of the eigenvalue of $M^T M$ (red) drops to 0 from $\varepsilon = 10^{-8}$ on

Figure 8 repeats the box for twelve values of ε . Watch the red line: it agrees with the blue one while σ_2^2 still fits in the 16 digits next to $\sigma_1^2 = 4$, then falls off a cliff. So `np.linalg.svd` never forms $A^T A$; NumPy calls the LAPACK routine `gesdd` (NumPy docs, `numpy.linalg.svd`). That routine reduces A itself step by step with orthogonal matrices, which do not magnify rounding errors, until the singular values can be read off (Trefethen and Bau, Lecture 31).

Extra: The ratio σ_1/σ_n of the largest to the smallest singular value is the **condition number** (G-441) of a matrix. It measures how much errors in the input can be magnified by solving with that matrix. Forming $A^T A$ squares it: a condition number of 10^8 becomes 10^{16} , which uses up all 16 digits (Trefethen and Bau, Lectures 12 and 19). The squaring is why the least-squares solvers of the [multiple linear regression code Note](#) (section 6) prefer to work from X itself rather than from $X^T X$. `np.linalg.cond(A)` computes it.

9. Summary

Step	What we compute	For A (rows $[3, 0]$, $[4, 5]$)	For C (rows $[2, 1]$, $[4, 2]$)
1	$A^T A$	rows $[25, 20]$, $[20, 25]$	rows $[20, 10]$, $[10, 5]$
2	eigenvalues $\rightarrow \sigma_i = \sqrt{\lambda_i}$	$45, 5 \rightarrow 6.708, 2.236$	$25, 0 \rightarrow 5, 0$
2	eigenvectors $\mathbf{v}_i$	$\frac{1}{\sqrt{2}}[1, 1]$, $\frac{1}{\sqrt{2}}[-1, 1]$	$\frac{1}{\sqrt{5}}[2, 1]$, $\frac{1}{\sqrt{5}}[-1, 2]$
3	$\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$	$\frac{1}{\sqrt{10}}[1, 3]$, $\frac{1}{\sqrt{10}}[-3, 1]$	$\frac{1}{\sqrt{5}}[1, 2]$
4	complete U	nothing missing	$\frac{1}{\sqrt{5}}[-2, 1]$

- $A^T A = V\Sigma^T \Sigma V^T$: eigenvectors are the $\mathbf{v}_i$, eigenvalues the σ_i^2 .
- $A^T A$ is always symmetric and positive semi-definite, so this always works.
- Get the $\mathbf{u}_i$ from $A\mathbf{v}_i/\sigma_i$, not from a separate eigen-problem: that keeps the signs matched.
- Zero singular values belong to the null space; the missing $\mathbf{u}$'s are any perpendicular completion.
- The SVD gives perpendicular bases for the row space, null space, column space and left null space.
- Computers compute the SVD from A directly, because $A^T A$ loses small singular values.

10. Sources

Built from

- 3Blue1Brown (Sanderson, G.), “Inverse matrices, column space and null space | Chapter 7, Essence of linear algebra”, YouTube, <https://www.youtube.com/watch?v=uQhTuRIWMxw>
- Khan Academy (Khan, S.), “Column space of a matrix”, YouTube, <https://www.youtube.com/watch?v=...>
- Khan Academy (Khan, S.), “Rowspace and left nullspace”, YouTube, <https://www.youtube.com/watch?v=...>
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11. Key terms

Term	Meaning
$A^T A$	The symmetric, positive semi-definite matrix whose eigenvectors are the right singular vectors and whose eigenvalues are the squared singular values
Row space	The span of the rows of a matrix; spanned by the $\mathbf{v}_i$ with $\sigma_i > 0$
Null space	All vectors a matrix sends to $\mathbf{0}$; spanned by the $\mathbf{v}_i$ with $\sigma_i = 0$
Column space	The span of the columns: every possible output; spanned by the $\mathbf{u}_i$ with $\sigma_i > 0$
Left null space	The output directions perpendicular to every column; spanned by the remaining $\mathbf{u}_i$
Rank (of a matrix)	The number of dimensions of the column space: how many dimensions the outputs fill
Four fundamental subspaces	Row space, null space, column space and left null space of a matrix
Condition number	σ_1/σ_n : how much a matrix can magnify errors when we solve with it